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Allergic Diseases in India – *prevalence, risk factors and current challenges*

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ABSTRACT

Epidemiological studies have shown a rise in the prevalence of allergic diseases in India during the last two decades. However, recent evidence from the Global Asthma Network (GAN) study has observed a decrease in allergic rhinitis, asthma and atopic dermatitis in children. Still, with a population over 1.3 billion, there is a huge burden of allergic rhinitis, asthma and atopic dermatitis and this is compounded by an unmet demand for trained allergy specialists and poor health service framework. There is wide variation in the prevalence of allergic diseases between different geographical locations in India and the reasons are unclear at present. This may at least in part be attributable to considerable heterogeneity in aero-biology, weather, air pollution levels, cultural and religious factors, diet, socioeconomic strata and literacy. At present factors enhancing risks and those protecting from development of atopy and allergic diseases have not been well delineated, although there is some evidence for the influence of genetic factors alongside cultural and environmental variables such as diet, exposure to tobacco smoke and air pollution and residence in urban areas.

This narrative review provides an overview of data from India regarding epidemiology, risk factors and genetics and highlights gaps in evidence as well as areas for future research.

Key words: *India, prevalence, asthma, rhinitis, atopic dermatitis, genetics, allergy*

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Key messages

- There are over 30 million people in India with asthma, often with poorly controlled disease
- Allergic diseases in India are exacerbated by high levels of indoor and outdoor air pollution
- There is a high rate of IgE sensitisation to foods with low food allergy rates

INTRODUCTION

India is experiencing an epidemiological transition with an increase in the burden of non-communicable diseases. Earlier studies suggested a rise in the prevalence of allergic diseases. However, the recent Global Asthma Network (GAN) study showed a reduction in the prevalence of allergic rhinitis, asthma and atopic dermatitis among children compared to the International Study on Asthma and Allergies in Childhood (ISAAC) studies. These changes in allergic diseasehave coincided with industrialisation, worsening air quality, changing dietary habits and other environmental factors,[1-4].[5] The aim of this narrative review is to describe changing trends in the prevalence of allergic diseases in India, associated risk factors, genetic studies in the native Indian population and challenges in the delivery of allergy services.

Methods

PubMed database was used to search studies using key terms for asthma, allergic rhinitis, atopic dermatitis, genetics, and food allergies as summarised in Table S1. For this narrative review, we considered multi-centre prevalence studies involving children and adults and for food allergy, smaller community studies were also included. Genetic studies from India mostly involved single centres.

1. Burden of allergic diseases in India

1.1 Airway allergic diseases

1.a Asthma

Children: The prevalence of asthma has risen from 1% in 1964 to 14% in 2017 in some geographical locations.[6,7] While physician awareness has increased and the diagnostic tools are more widely available, this steep increase cannot be explained by these factors alone. However, the recent multicentre Global Asthma Network (GAN) study observed a reduction in the prevalence of asthma from the ISAAC III to GAN I.[5,8] A wide variation in asthma prevalence has been reported in different regions of India ranging from 2% to 23% among school children in ISAAC (2003) and 1-20% in the Global Asthma Network study (2019).[1,2,5,8-10] A wide variation in the prevalence of asthma is seen in the rest of the world as well. The highest burden of childhood asthma (GAN study) was seen in Costa Rica and New Zealand in 6-7 year

olds and South Africa, Costa Rica and Syria had the highest burden among 13-14 year old children.[11] The prevalence of asthma in the USA was 12.5% & 11.9% and Europe 11.2% & 15.2% among 6-7 and 13-14 year olds respectively.[11] Before the GAN study showed a declining trend, the prevalence of asthma had risen not only in metropolitan cities but also in Indian lower-tier cities and rural areas.[2,3,7,12,13] **Figure 1** summarises asthma prevalence among children aged 6-7 years and 13-14 years in different Indian locations, both from the ISAAC and the GAN studies.[5,8,11,14]

Adults: Nationally representative samples from all the states of India were included in the National Family Health Surveys (NFHS). The database from NFHS-3, recorded the prevalence of self-reported asthma in adults to be less than 2% in 2005-2006, and a similar rate was seen in the NFHS-2 between 1998-1999.[7,15] Similarly, the Indian Study on Epidemiology of Asthma, Respiratory Symptoms and Chronic Bronchitis in adults (INSEARCH) conducted in 12 districts in different parts of India reported asthma prevalence of 2.05% in those aged ≥15 years between 2007-2009.[4] Recent data from NFHS-4 from all the states of India reported the prevalence of asthma to be 2% and 1% amongst women and men respectively in the 15-49 years age group and the prevalence of asthma among adults was similar from 2005-2006 until 2015-2016.[16] However, data from Indian Human Development Survey revealed an increase in asthma prevalence from 41.9 per 1000 population in 2004-2005 to 54.9 per 1000 population in 2012-2013, although the number of households surveyed were relatively low and the survey design was somewhat different from NFHS.[15] **Figure 2a** describes the adult asthma prevalence in 2019 in different Indian states and union territories according to the Global Burden of Disease (GBD) study.[17] In comparison, the prevalence of asthma in adults in USA (NHANES study) was 6% and in Europe (ECRHS) was 2-11%. [18,19]

1.b Allergic rhinitis

Children: The prevalence of AR had risen in India between the two ISAAC studies. An increase in AR was observed amongst children aged 13-14 years from 18.6% to 23.6%, and in children aged 6-7 years from 12.5% to 12.9%, in the ISAAC phase I and phase III studies.[20,21] In the recent GAN study, a reduction in the prevalence of allergic rhinitis was observed in 6-7 year olds (7.7%), while the prevalence was similar among 13-14 year olds (23.5%). Wide variation in allergic rhinitis prevalence has been reported in different regions of India ranging from 0.9% to 26.7% amongst school children in ISAAC (2003) and 7.1% to 24.1% in the GAN study (2019).[22,23] Wide variations in allergic rhinitis prevalence was also observed in the rest of the world. The highest burden of allergic rhino-conjunctivitis (ISAAC III) was seen in Taiwan (24.2%) and Hong Kong (17.7%) in 6-7 year olds and Morocco (28.1%) and Chile (26.3%) had the highest burden among 13-14 year old children. The prevalence of asthma in USA was 6.4-10.8% & 19.1% and Europe 10.1% & 15.3% among 6-7 and 13-14 year olds respectively.[24] **Figure 1** summarises AR prevalence amongst children aged 6-7 years and 13-14 years in different Indian locations from the ISAAC and the GAN phase I, India study.[14,22,25]

Adults: The INSEARCH study (2012) involving 169,575 subjects, reported a prevalence of 3.45% for AR in Indian adults.[26,27] The GAN adult study observed an

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increase in prevalence at 9.8% (2019).[22] Allergic rhinitis is the most frequent co-morbidity in asthma and a recent survey of asthmatics from urban areas showed a prevalence of 65.4%.[28,29] In comparison, AR prevalence in US adults was 41.2% and in Europe (ECRHS I) 28.9% .[30].

1.2. Atopic Dermatitis

Children: In India, the ISAAC phase III study in children aged 6-7 years showed 'current prevalence' of AD and AD 'ever' as 2.7% and 4.4% respectively; 3.6% and 8.9% for age 13-14.[31] ISAAC phases I and III showed an increase in AD prevalence in Indian children.[24,32] The GAN study (2019) observed 2.5% prevalence in 6-7 year olds and 3.5% in 13-14 year olds. The highest burden of AD (ISAAC III) was seen in Sweden (20.9%) and UK (14.5%) in 6-7 year olds and Ethiopia (19.4%) and Morocco (28.1%) among 13-14 year olds. **Figure 1** summarises AD prevalenceat 6-7 years and 13-14 years in India from ISAAC and GAN.[14,22]

Adult: Global Burden of Disease (GBD) data show increased adult AD prevalence at <50 years (1.46%) compared with >50 years (0.47%) and higher prevalence in females for both age groups.[17] **Figure 2b** illustrates adult AD prevalence in India from GBD.[17] In comparison, AD prevalence in US adults (NHANES, 2016) was 6.2% and in Europe (2018) 4.4% (95% CI: 4.2%, 4.6%).[33]

1.3. Food allergy

Children: Interestingly, food allergy is not a major public health problem in India. The EuroPrevall study is the largest population-based study on food allergy and included India, China and Russia. Overall 5677 children aged 7-10 years were screened using questionnaire, skin prick tests and specific IgE to 25 foods, and 450 participated in the detailed case-control study. Sensitisation to any food using specific IgE was found in 19.1% in India, with shrimp (10.5%) and sesame (8.0%) most common. Using skin prick testing the rate was 4.48%, with jackfruit (2.46%) and milk (1.35%) most common. 'Probable food allergy' was defined as a history of clinical symptoms ≤2hours after consumption of a food with a positive specific IgE or skin prick test to the same food. In Europrevall only 0.14% of children in India had 'probable food allergy', with cow's milk (0.5%), apple (0.5%), egg (0.05%) and eggplant (0.04%) most common.[34] Very few studies in India have confirmed food allergy with a challenge procedure. Kumar et al reported 40 hospital based open food challengeswith confirmed allergies to rice (6%), blackgram (5.9%), lentil (5.5%) and citrus fruits (5.3%).[1,35] The same group conducted double-blind placebo controlled food challenge (DBPCFC), in patients sensitised to rice and black gram and confirmed true food allergy to rice (6/16) and blackgram (4/14).[35]

Adults: The EuroPrevall-INCO survey in two South Indian cities revealed self-reported adult food allergy prevalence of 1.2% but 26.5% were sensitised.[34,36] Legumes, prawn, eggplant, milk and egg were the main food allergens.[37] The low prevalence of food allergy in India has been attributed to multiple factors such as relatively longer breast-feeding, vaginal delivery, traditional vegetarian diet, diverse gut microbiota and enteric helminthiasis.[38,39] The recent shift from the traditional Indian lifestyle to westernisation could potentially alter natural protective factors and lead to enhanced sensitisation rates to food allergens. However, reasons

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underpinning relatively low rates of clinical reactivity in sensitised subjects remain uncertain.

1.4 Drug Hypersensitivity

There is a paucity of information on the incidence of drug hypersensitivity in India. There is a need for good pharmacovigilance programs, currently unavailable in most hospitals in India.[40] A systematic review reported the pooled incidence of drug hypersensitivity among outpatients and inpatients was 9.22/1000, most commonly maculopapular rash (32.39%), fixed drug eruptions (FDEs) (20.13%) and urticaria (17.49%). Severe reactions such as Stevens-Johnson syndrome/toxic epidermal necrolysis (SJS/TEN) were seen in 6.84% of subjects who developed adverse drug reactions (ADRs), most commonly due to antimicrobials (45.46%) such as sulphonamides and β-lactams, nonsteroidal anti-inflammatory drugs (NSAIDs) (20.87%) and anti-epileptic drugs (14.57%) such as carbamazepine (6.65%). Mortality for severe cutaneous drug reaction and exfoliative dermatitis was 1.71% and 3.57%, respectively and 16.39% for SJS/TEN. “Definitely preventable”, “probably preventable” and “not preventable” categories ADRs were 15.64%, 63.14%, and 34.64%, respectively.[41]

2. Patterns of sensitisation to aero-allergens and food allergens:

India is a land of diversity with heterogeneity in flora, fauna, climate, lifestyle and food habits. It is estimated that >25% of Indian population are sensitised to one or more allergens.[42] Pollen grains, fungal spores, dust mites, insects and food are the important allergen sources.[26] Since India has various eco-zones, the pattern of allergic sensitisation to various pollen grains also varies between different geographical areas. From Northern India, *Prosopis juliflora, Ricinus communis, Morus alba, Mallotus phillipensis, Alnus nitida, Quercus incana, Cedrus deodara, Argemone mexicana, Amaranthus , Chenopodium album, Holoptelea Integrifolia,* and grasses constitute the important sensitising pollens.[43] In Central India the most common are *Argemone mexicana, Brassica oleracea, Cannabis sativa, Parthenium hysterophorus, Cassia siamea, Azadirachta indica* and grasses.[44] *Lantana camara, Cucurbita maxima, Cassia fistula, Cocos nucifera, Callophyllum inophyllum, Phoenix sylvestris , Ricinus communis and Aegle marmelos* are the important pollens from Eastern India.[26] In South India *Cassia spp (occidentalis & siamea), Ageratum conyzoides, Salvadora persica, Ricinus communis, Albizia lebbeck, Artemisia scoparia, Parthenium hysterophorus* and *Casuarina equisetifolia* are the major allergenic pollens.[45,46] *Pinus roxburghii* sensitisation is seen in 16.9% of patients in the Himalayan foothills. Under the All India Coordinated Project (AICP) on aero-allergens and Human Health, 43 types of dominant pollens are recognised with *Holoptelea Integrifolia, Poaceae, Eucalyptus Myrtaceae, Melaleuca quinquenervia, Casuarina equisetifolia, Putranjiva roxburghii* having pan-India relevance (**Figure 3).**[1]

AICP reported *Aspergillus ochraceus, Aspergillus japonicus, Cladosporium asperulatum, Alternaria alternata, Aspergillus versicolor, Uromyces* and *Ustilago trtici* to be the major allergenic fungi. Tropical climatic conditions contribute to the growth and spread of fungal spores and allergenic sensitisation.[37] House dust mite (HDM) and cockroach sensitisation rates vary from 12.4% to 82% and 29.89% to 37.86%

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respectively amongst Indian patients with naso-bronchial allergy.[44,47] Varying sensitisation rates have been attributed to differences in building materials, hygiene standards, socioeconomic status and microclimatic eco-systems [9]. Sensitisation to Dermatophagoides *pteronyssinus* is higher than Dermatophagoides *farinae* in most regions and in West Bengal sensitisation to *Blomia tropicalis* is seen.[48] Sensitisation to fungi was higher in younger subjects from rural area and cockroach sensitisation was higher in younger subjects from urban areas.[49]

The common food allergies in India are shellfish, legume, milk, eggand eggplant allergy.[50] As described above, although food sensitisation rates are up to 19.1% in children and 26.5% in adults, clinical reactivity is uncommon (0.14% children, 1.2% adults).[36] Studies from various parts of India implicated chickpea, black gram, lentils, prawn, banana, egg, fish, milk, peanut, wheat and apple as important sensitising food allergens.[51]

Comparison between India and other LMIC countries

The general population in India has high rates of sensitization and low clinical reactivity.[34,36] Sensitization and clinical disease in urban areas are higher than in rural areas.[52] It is useful to compare this with observations from other LMIC countries.

In Indonesia, urban children of lower socio-economic status (SES) had more wheeze and asthma, high total IgE (>10,000 IU/ml) and heminth infection (>90%) but low positive skin prick test rates (lower SES 8% vs higher SES 25%) for HDM and cockroach.[53] In Latin American rural children lower SES and helminth infections were associated with lower atopy.[54]Several studies in Africa (Ethiopia, South Africa, Kenya, Gambia) have observed lower allergic diseases in ruralthan urban areas.[55] In the South African Food Allergy (SAFFA) cohort,[56] clinical allergies and sensitization to food and aeroallergens were all higher in urban than rural children. In Indonesia, rural children had less asthma and skin prick test positivity to HDM and cockroach, but higher total IgE (>2 fold) and higher specific IgE to HDM and cockroach than semi-urban children.[57] In Subaraya, Indonesia sensitization rates increased with age; primary school (21.9%), secondary school (28.9%), and undergraduate (48.3%).[58] The prevalence of asthma in these students was 6.8% and food allergy 3.4%. In Makassar, Indonesia, 24.6% of low-SES children had sIgE for HDM and cockroach, and 20.7% of high-SES children.[53] However, skin prick test positivity was lower among low-SES (5.7%) than high-SES (16.4%) and asthma was similar (7% high-SES, 6.1% low-SES). An interesting aspect of this study was the observation that children with poorer nutritional status had less sensitization.[59] Aeroallergen sensitization rates among children from other LMIC's range from 8.7% in Ethiopia, to 25% in Uganda.[60] In SAFFA, food sensitization and allergy were higher in urban (9.0%, 2.3%) than rural areas (2.8%, 0.5%). Very low food sensitization rates were observed in Uganda (<1.5%), Equador (0.5%) and Indonesia (0.8%).[60] The differences between sensitization rates to foods and food allergy in other LMICs are not as stark as that reported from the Indian population.

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3. Key determinants of allergic diseases in India

3.1 Airway allergic diseases

3.1a. Asthma: ISAAC study reported key determinants associated with asthma in 6-7 year olds as maternal smoking, sedentariness, paracetamol in infancy, indoor pollution; and maternal smoking, outdoor pollution, male gender and underweight for 13-14 year olds.[5,8,14,61] GAN study reported reduced fresh fruit consumption and reduced breastfeeding to be important.[22] The reduced asthma prevalence in GAN may reflect the reduced number of children exposed to air pollution through residing in areas where trucks pass frequently.[25]

3.1b. Allergic rhinitis: ISAAC study observed increased AR at age 6-7 years with maternal smoking, paternal smoking, sedentariness, paracetamol intake, outdoor and indoor pollution, male gender, underweight, obesity, damp and premature birth; andmaternal smoking, outdoor and indoor air pollution, male gender, underweight, obesity at age 13-14.[25] GAN reported damp, premature birth, paracetamol and antibiotics and laying on sheep skin in infancy and pneumonia at age 6-7 years were associated with AR; and sedentariness, recent paracetamol, use of water coolers, banana, cold drinks and fast foods, at age 13-14 years.[22]

3.2 Atopic dermatitis:

ISAAC found maternal and paternal smoking, sedentariness, current paracetamol were associated with AD in both age groups. Indoor and outdoor air pollution was associated with AD at 6-7 years and indoor pollution at 13-14 years.[5,25] GANfound antibiotics, farm animal exposure during pregnancy, air conditioners antibiotic in infancy, meat consumption, sedentariness, cat exposure, fast food consumption, outdoor pollution and recent paracetamol were associated with AD .[22] (**Figures 4 and 5)**[14,22,25]

3.3 Food allergy

Food allergy prevalence is too low in India for reliable analysis of risk and protective factors. Other data suggest early introduction of common allergenic foods is protective, and other dietary and non-dietary factors have been proposed.[62]

3.4. Specific influenceson allergic diseases:

Environmental tobacco smoke(ETS): ETS exposure in children is an important modifiable risk factor for childhood asthma.[63] Maternal smoking has greater asthma impact than paternal smoking.[64,65] However in ISAAC III paternal smoking had a greater effect in Indian centres, probably due to low maternal smoking during pregnancy.[5,8] .[61] ISAAC reported a dose response increase in childhood asthma for both maternal and paternal smoking, also seen for paternal urinary cotinine in Korea, and for paternal smoking and DNA methylation of immune genes in Taiwan.[66] .[67].[68]

Area of Residence: Urban residents, including children have more asthma rural residents.[69] Possible reasons include house dust mite, vehicle emissions, sedentariness, ETS and diet.[7071] [71-74] [74]

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In a South Indian study, aeroallergen sensitisation was six times more prevalent in urban than rural children who had close contact with cow dung and animals and five times higher endotoxin levels[52] An inverse association between endotoxin exposure and sensitisation or asthma has also been reported outside India.[75]

Air pollution: Indoor air pollution from cooking fuels, incense, mould and cleaning materials is associated with increased asthma.[68] [22,63] Both indoor pollution and outdoor pollution from traffic and industry are associated with asthma, AR and AD.[61] [22]

Climate change: This is likely to have a major impact on asthma and AR, through increased pollen count, dispersal and allergenicity, but more studies are needed.[76,77].[7878,79] Airborne particulate matter and ozone interact with pollen to promote sensitization.[78] Increasing global temperature and carbon dioxide levels increases pollen production.[80] Changing temperature and rainfall patterns have influenced the phenology pattern in India.[81-83] Extreme weather conditions such as cyclones, heavy monsoons, floods cause increased indoor moulds and microbial growths, in turn aggravating allergic diseases.[84]

Obesity: Childhood obesity is a risk factor for childhood asthma.[85-88] Increased IL-6, IL1β, TNFα and leptin may influence pulmonary response to environmental triggers.[89,90] Shared hereditary factors for asthma and obesity may include PRKCA, LEP, DENND1B and ADRB3 genes, although, further studies are needed.[89]

Medications: Paracetamol in early life is associated with asthma, AR, and AD.[91] Low pulmonary glutathione and oxidative stress are possible mechanisms affecting FEV1.[91] Frequent antibiotic exposure in children may promote allergic diseases and mode of delivery may be relevant for asthma.[9293] Antibiotics in early life is also associated with increased allergic diseases in India.[8,22]

Sedentariness: sedentary lifestyle is associated with central obesity, a risk factor for allergic diseases.[94] In India, sedentary behaviour is associated with increased allergic disease.[22] Sedentary behaviour and lower aerobic fitness are associated with increased airway inflammation and lower regulatory T cell responses.[95]

Gender: ISAAC found male gender was a risk factor for children allergic diseases in both age groups. This may be due to hormonal or immunologic factors.[49,50] Boys may also have increased bronchial reactivity [3,96].[9]

Mode and timing of delivery: Caesarean section (CS) is associated with asthma with relative risk 1.18 (95% CI 1.07, 1.29; p<0.001).[97] About 78.5% of women in India have normal vaginal delivery.[97,98] Overall, studies of mode of delivery are inconsistent, but altered microbiome, and lack of stress hormone affecting the growth and development of lungs, may be relevant.[99,100].[100,101] Premature birth is an important risk factor for asthma [OR= 1.92; 95% CI (1.23, 2.99) p<0.001],[102].[103]

Breast feeding and diet: Breastfeeding practices showed no association with asthma, AR and AD,[20] unlike studies in other ethnic populations (Denmark and Japan)[104,105] which showed varied observations[105] .[20] High dioxin levels in breast milk may affect the development of allergic diseases.[104] Exclusive breastfeeding

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(EBF) in India is around 63.7% (NFHS, 2021),has increased overall (NFHS, 2005) and is higher in South India and rural areas.[98 106107]

Other aspects of diet vary across India. [108] With a few exceptions (Kerala, West Bengal, Meghalaya), non-vegetarian food is not consumed regularly.[109] Indians have a highly diverse diet with many complex ingredients including spices and there are no studies evaluating dietary diversity and prevalence of asthma. The European Protection against Allergy STUdy in Rural Environments (PASTURE) study observed that higher dietary diversity was associated with lower asthma.[110]

Gut microbiome: In a large multicentre study of gut bacterial characteristics across 14 centres in India, 993 unique microorganisms were observed.[111] Only 390 microorganisms were common to all geographical locations and Prevotella copri and Faecalibacterium prausnitzii are the predominant microorganisms.[111-114] Prevotella genera predominantly occurred among vegetarians with a plant-based diet and high fibre consumption.[113,115] Bifidobacterium longum and Lactobacillus acidophilus predominated in children under 3, and Prevotella-Porphyromonas, Eubacterium. rectale and Faecalibacterium prausnitzii in older children and adolescents.[116] There are no studies in India of microbiome and allergic disease, but other studies have observed negative associations with relative abundances of genera Faecalibacterium, Bifidobacterium, Ruminococcus, and Dialister and a positive association with Veillonella and asthma.[117]

3.5 ADULTS

For adults in the INSEARCH study, a family history of asthma, tobacco smoke and environmental tobacco smoke, female gender, urban residence and indoor air pollution were identified as key determinants.[4]

Figure 6 illustrates risk factors contributing to allergic diseases in adults from the INSEARCH and GAN phase I study. Mould, smoking and mosquito coils were associated with AR. and smoking and mosquito coils with AD.[22] Specific foods and air conditioners were protective against AD and meat consumption against AR.[20]

Family history of asthma was the strongest risk factor and it is likely that 40-60% of asthma is caused by genetic factors. Several gene-gene interactions and gene-environment interactions are crucial in establishing asthma in a given subject.[118] Smoking is associated with increased asthma and a dose response relationship was seen in Canada.[119] In adults, asthma is increased in women, perhaps due to female sex hormones.[120]Gender influences gene polymorphisms effects. In at least six different ethnic populations, certain polymorphism (T allele of rs1837253) of thymic stromal lymphopoeitin (TSLP) was associated with reduced asthma only in men whereas another polymorphism (T allele of rs2289276) in TSLP was associated with reduced asthma in women only.[121] Gender may differentially modulate the transcription factor binding site by particular TSLP polymorphisms.[121] In female mice, higher TSLP had eosinophil dominant perivascular inflammation compared to males and gender-specific anti-oxidant responses were also seen.[120]

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4. Genetic studies in asthma from India

Candidate gene studies in India are summarised in Table 1. The established risk factor *ADAM33*, was confirmed in both North and South Indian populations.[122-124] Polymorphisms in transcription factors *TBX21* and *FOXO3a,* are associated with asthma development and severity. [125,126]

In an Indian study of 33 Th2 relevant genes population, SNPs in *IRF2, IFNγR2, IL6, STAT4 and IL4RA*, were associated with asthma.[127] Other studies have reported polymorphisms in immune genes for *IL4*, *IFN□, IL7R□1, IL13R, TIM1, IKAP and* Mannose Binding Lectins *(MBL)* associated with asthma or total IgE.[128-136]

Monocyte-macrophage pathway studies found polymorphisms in the *MCP3*, *CCR2*, *TLR4*, and *CD14* genes associated with asthma or total IgE.[137-140] Studies investigating mediators of immune resolution identified microsatellite repeats in *CC16*, haplotypes on the *IL10* promoter and novel (CT) (CA) repeats in the *TGFβ* promoter associated with asthma.[141-143] SNPs on *IL10* and *IL17F* promoters, CHIA, CHIT1, CHI3L1 and LTCS genes were associated with disease phenotypes.[144] [145-148]

Polymorphisms in genes linked with IgE eosinophil axis are associated with asthma e.g. exon 3 of *IL21*.[149] Polymorphism in the*FcεRIβ,* mast cell chymase gene (*CMA1*) and *eotaxin* was associated with asthma severity and serum total IgE levels.[150-152] Polymorphisms in N-acetyltransferase 2 were associated with total IgE levels and eosinophil counts.[153] Polymorphisms in genes involved in signalling mechanisms like INPP4A, and those involved in metabolism like the vitamin D receptor (VDR), were also associated with asthma.[154,155]

Genome Wide Association Studies (GWAS) for asthma conducted in India are limited, but whole genome and exome sequencing identified association of asthma with a missense variant in the olfactory receptor gene (*OR2AG2*).[156]

Epigenetic data from North India highlighted the effect of expression of some of these proteins [corticosteroids- P glycoprotein (P-gp); Histone Deacetylase 2 (HDAC2)] on asthma control.[157] An animal model based study from New Delhi has recently pointed towards control of mast cell activation by newer molecules [Fc epsilon RI (FcεRI)] which could serve as targets for newer drugs.[158]

A major limitation of GWAS is its inability to effectively negate the heterogeneity seen in the population or the disease. This could be minimised by stratification of the sample populations according to age, ethnicity and disease endotypes according to severity or pathogenesis. Recent reports highlighted the overlapping interface between systemic diseases such as metabolic syndrome and asthma. Although initially thought to be a consequence of systemic inflammation, the obese-asthma phenotype is now believed to have other underlying causes such as insulin resistance [159,160]. Literature has also emphasised the role for environmental factors, especially cigarette and traffic smoke exposure in influencing gene regulation in multifactorial diseases such as asthma.[161-164] Unbiased genetic studies would form an important arm towards identification of such intersecting mechanisms, which remain unexplored. The genetic

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differences in the ethnic population in India suggest that environmental (indoor and ambient air pollution) factors and diet influence urban and rural populations. Overall, among 14 cities in the ISAAC study, nine cities had an air quality index (AQI) of less than 100 and the rest more than 100. Thus air quality index (AQI) across regions and geo-climatic differences would influence the prevalence of allergic airway diseases.

Understanding evidence gap

The earliest published work from India was on respiratory allergy to moulds in 1961. Six decades have elapsed, but advancement in allergy and immunology has been relatively modest. Birth cohort studies in allergy and asthma are lacking, data are sparse regarding time trends and prevalence of AR and reasons for low AR prevalence among adults in India are unclear. The co-existence of asthma and AR however has been reported amongst Indian patients in few studies[29,34]. The evaluation of ethnicity and genetics, cultural and religious influences, and socioeconomic factors and their impact on asthma and allergy has not yet been fully elucidated in the Indian population. Disease management is largely based on international guidelines such as Global Initiative for Asthma (GINA) and Allergic Rhinitis and its Impact on Asthma (ARIA). The impact of biodiversity and environmental change on respiratory allergy has not yet been addressed in a concerted manner in the native Indian patients. Moreover, the data on the patterns of aero-allergen distribution in India is two decades old. Regular reports on aero-allergens, updated pollen calendars for various regions of the country are needed to facilitate management of pollen allergy and for establishing fit for purpose allergen-specific immunotherapy protocols. Data on food allergy from India is also limited to EuroPrevall-INCO study, where only two centres from South India participated. Similar datasets are needed from other regions of India. Breast feeding practices and dietary intake, influence the gut microbiota among children, especially in urban areas.[165] Additionally, a significant number of people consume food from the street vendors which introduces a substantial amount of cross contamination of allergens and an enhanced risk.[30] The C-section rates are increasing in India[98] (21.5% NFHS, 2021) and hence there is an additional need to better understand the impact of this procedure on gut microbiome and risk of asthma and allergy. Similarly, data on drug allergy, stinging insect allergy and anaphylaxis are sparse in India. Nationwide disease registries are not yet available at present for these disorders as in the UK and Europe.

Practical considerations for delivery of allergy and asthma care in India

There is huge demand for an allergy services and postgraduate training in asthma and allergy. ~37.9 million people with asthma live in India.[166,167] Prevalence studies suggest early 20% of the population (~200 million) have at least one form of allergic disease.[167] The POSEIDON study reported respiratory symptoms as the most

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common clinical presentation to a physician in India, comprising half of all cases and two-thirds of children, withobstructive airways disease one of the most common diagnoses.[168]

Awareness of allergic disorders especially among paediatricians has increased due partly to the Indian Academy of Paediatrics forming a separate chapter dedicated to education of the membership. Nearly 30-40% of outpatient paediatric practice is related to allergic disorders .[169] It has also been recognised by phenotype clustering that childhood onset asthma is a major component of the disease burden in India.[167]

In response to the demand for specialisation in allergy, the first allergy short education program at Vallabhbhai Patel Chest Institute, in collaboration with CSIR-Institute of Genomics, New Delhi, began in 1975 and has trained more than 1500 doctors from different specialities in asthma and allergy management. The 'Diploma in Allergy and Asthma (DAA)' was initiated at Christian Medical College, Vellore, Tamil Nādu in 2006 in partnership with International Asthma Services (IAS) (www, globalchestinitiatives.org).[170] DAA is sponsored/endorsed by the American Academy of Allergy, Asthma & Immunology (AAAAI) and the Centre for Global Health, University of Colorado, USA. Nearly 500 physicians have graduated from the DAA program and some DAA alumni have started additional training programs in New Delhi, Kolkata and Bangalore.[169] The first online teaching program (with onsite practical training) in India namely Allergy and Asthma Specialist Course (AASC) is a unique course offered in Bangalore in partnership with Med-Train, a local medical online educational service and IAS, Colorado USA and is also endorsed by AAAAI.

It is estimated that an allergy workforce might ideally be 1:50,000 population. Extrapolating this to the Indian scenario, there would be a need for 20,000 allergy specialists. At present there are an estimated 1000 allergy and asthma specialists, a ten-fold increase over the last 3 decades. It is estimated that the demand for allergy training programs will continue to grow and will evolve substantially in the next decade. The ideal situation in India is to establish a formal postgraduate training programme for physicians exclusively in the specialty of allergy, asthma & clinical immunology similar to the 2-year fellowships in North America. Such a course needs to be recognised by the Indian Medical Council. There is a need to train healthcare providers in primary care and secondary care in the management of allergies and asthma and for establishment of referral pathways.

<u>Access to investigations and medications</u>: There is lack of standardised allergen extracts, particularly house dust mite , which is the most common allergen in India.[70]Access to *in vitro* allergy testing is challenging with multiple barriers such as unaffordability, limited access and lack of quality assurance programmes. Serum specific IgE measurement is undertaken on ImmunoCAP platform in larger laboratories. However, ImmunoCAP is not widely available and may not include locally relevant indigenous aeroallergens in India. Non-standardised methods such as ELISA are widely used by some laboratories with a limited repertoire of allergens.

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The lack of availability of adrenaline auto-injectors is another important barrier in the management of patients at risk of food, insect sting allergies and spontaneous anaphylaxis. At present, patients carry pre-drawn injectable adrenaline in a tuberculin syringe secured in a spectacle case or a box.

There is also a need for patient education resources for a large and diverse population with respect to multiple languages, religion, culture, deprivation and literacy.

National guidelines and resources for healthcare professionals and patients: There is a need for development of national guidelines for asthma and common allergic conditions targeted to the Indian healthcare framework, encompassing public, private and corporate sectors. One guideline on immunotherapy has been published recently.[171] Similar guidelines have to be developed for the diagnosis and management of allergic diseases in India. Provision of educational resources for patients and their carers might enhance uptake of management plans and improve clinical outcomes, particularly for asthma.

FUTURE DIRECTIONS

A systematic characterisation of allergic disease phenotypes, endotypes and disease clusters is needed in the native Indian population. This would facilitate translational research and clinical trials, specifically with respect to appropriate use of immunomodulatory and biologic therapies. At present clinical practice is largely based on data generated from high-income countries in largely Caucasian White populations. Parameters such as serum total IgE, peripheral blood eosinophil counts and sputum eosinophil count are likely to be different in India due to ethnicity, parasitic and other infectious diseases and other environmental factors including air pollution.

Like other regions, there is a need for studies on impact of climate change on allergic diseases. Investigating factors protecting and enhancing risk of allergic diseases in regions of India and in Indian migrant populationsis likely to pave the way for primary and secondary prevention strategies..

CONFLICT OF INTEREST

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AUTHOR CONTRIBUTION

PAM, DJCand MTK conceptualised and designed the work. SMalamardi extracted the data and performed the data analysis. SM, UM, MG, SMalamardi , PKV and AA drafted specific sections of the manuscript. PAM and MTK amalgamated and edited contributions from co-authors. All the authors reviewed and approved the final version of the manuscriptand agreed for submission.

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Table 1. **Summary of various genetic studies in the Indian sub-continent**

OR= Odds ratio; CI= Confidence Interval

Gene Name	Polymorphism	Population Studied	Estimates	Author, Year	Study Design
ADAM33	rs528557 rs511898 rs44707 rs597980 rs2787094 rs2280090 rs2280091	North and South Indian	0.03-<0.001	Awasthi et al., (2011)[122]; Tripathi et al., (2011)[123]; Vishweswaraiah et al., (2019)[124]	Case-control
TBX21	rs4794067	North Indian	0.001	Sharma et al., (2014)[126]	Case-control
FOXO3a	rs13217795	Western Indian	<0.0001	Barkund et al., (2015)[125]	Case-control
IRF2	rs3733475	North and North West Indian	0.002	Kumar et al., (2015)[127]	Case-control and Family
IFNGR2	rs2012075	North and North West Indian	0.004	Kumar et al., (2015)[127]	Case-control and Family
IL6	rs2069832	North and North West Indian	0.001	rs2069832	Case-control and Family
STAT4	rs1400656	North and North West Indian	0.003	Kumar et al., (2015)[127]	Case-control and Family
IL4RA	rs1805011	North and North West Indian	0.001	Kumar et al., (2015)[127]	Case-control and Family
IL4	CA repeat polymorphisms	North Indian	<0.05	Nagarkatti et al., (2004)[129]	Case-control
IFNγ	CA repeat polymorphism; rs1861494	North Indian	0.0006	Kumar and Ghosh (2008)[132]	Case-control and Family
IL7R	rs1494555 rs6897932	North Indian	0.02 - <0.001	Sinha et al., (2015)[131]	Case-control
IL13R1α	+1398A/G	North Indian	0.043	Sinha et al., (2015)[130]	Case-control
MCP3	-1382T/C (rs3091318)	North and North West Indian	0.002 - 0.00001	Batra et al., (2011)[138]	Case-control and Family

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CCR2	rs3918356 rs743660	North, North-West and North-East Indian	0.03 - 0.00005	Batra and Ghosh, (2009)[137]	Case-control and Family
TLR4	C>1196T	North Indian	0.046-0.039	Sinha et al., (2014)[140]	Case-control Study
CD14	C-159T functional promoter polymorphism	North and North West Indian	0.01-0.004	Sharma et al., (2004)[139]	Case-control and Family
CC16	microsatellite repeats, A_18 Haplotypes	North Indian	0.016 & 0.001	Sharma and Ghosh, (2004)[143]	Case-control and Family
IL17F	rs1889570	South Indian	0.004	Raeiszadeh Jahromi et al., (2015)[144]	Correlation between the SNP and allergens
IL10	ATA, GCC, ATC Haplotypes; rs1800871; rs1800896	North and South Indian	0.03 - <0.0001	Chatterjee et al., (2005)[141]; Jahromi et al., (2005)[144]	Case-control and Family
TGFβ1	$(CT)_n(CA)_m$ repeat polymorphism; 23_G_T; 22_G_C haplotype	North and West Indian	<0.05	Nagpal et al., (2005)[142]	Case-control and verified in 2 cohorts
CHIA	rs3806448 rs2282290	North Indian	0.00001-0.0002	Chatterjee et al., (2008)[146]	Case-control and verified in a pediatric cohort.
LTC4S	rs3776944	North Indian	0.009 & 0.0004	Kumar et al., (2012)[147]	Case-control and Family
IL21	C5250T polymorphism; 4_C_T_C_C; T_C haplotypes	North and North West Indian	0.01-0.0007	Chatterjee et al., (2009)[149]	Case-control and Family

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CMA1	$(TG)_n$ and $(GA)_m$ polymorphism; -1903 G/A; G_43 haplotype	North and West Indian	<0.05	Sharma et al., (2005)[152]	Case-control (2 independent cohorts)
Eotaxin	+67G/A; –384C/T polymorphism; $(GAAGGA)_n$; G_A_C_8 haplotype	North and North West Indian	0.009 & 0.006	Batra et al., (2007)[150]	Case-control and Family
FcεRIβ	A_16_C; G_18_T haplotypes	North and West Indian	10-5 & 0.004; 0.007 & 0.0004	Sharma et al, (2004b)[151]	Case-control (2 independent cohorts)
NAT2	C282T; C481T polymorphism	North Indian	0.001-0.0005	Batra et al, (2006)[153]	Case-control
INPP4A	1110832A/G	North Indian	0.004	Sharma, (2008)[154]	Case-control and Family
VDR	rs2228570	South Indian	0.022	Rajaram, (2019)[155]; Kaur (2006)[133]	Case-control
MBL	G1011A	North and South Indian	<0.01	Kaur et al., (2006)[133]; Suman et al., (2006)[136]	Case-control
CHI3L1	rs4950928	North Indian	<0.001	Naglot et al., (2015)[148]	Case-control
CHIT1	24bp duplication	North Indian	0.002	Sinha et al.,(2014)[145]	Case-control
IKAP	T3214A	North Indian	0.04	Sinha et al., (2016)[135]	Case-control
TIM1	rs41297579	North Indian	0.029	Sinha et al., (2015)[134]	Case-control
SP-D	T341C	South Indian	0.003	Suman et al., (2006)[136]	Case-control
ADRB2	GGCTTTGCAA haplotype	North Indian	0.021	Bhatnagar, (2005)[172]	Case-control

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FIGURES LIST

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FIGURES (TITLES and LEGENDS)

FIGURE 1 Prevalence of Asthma, Allergic Rhinitis, and Atopic dermatitis in different cities for age groups A) 6-7 years and B) 13-14 years

The figure shows the city-wise prevalence (as %) in children with a) Asthma (b) allergic rhinitis, and c) Atopic dermatitis. In the plot, navy blue represents children of age group 6-7 years and green represent age group 13-14 years from the ISAAC III study(represented as *); Orange and red represents 6-7 years and 13-14 years data from the GAN study (2022). Figures have been independently developed from the publicly available database of the ISAAC III study [Accessed on 8th May, 2021 at http://doi.org/10.5255/UKDA-SN-8131-1], GAN phase I, worldwide study (2021) in reference [https://doi.org/10.1016/S0140-6736(21)01450-1] and GAN phase I, India study (2022) in reference 20 [DOI: https://doi.org/10.1016/j.jacig.2022.01.004]

FIGURE 2A Prevalence of Asthma in adults across all the states and union territories in India.

The left panel show the prevalence (percent) of asthma in adults among A) the age group less than 50 years and higher age group; in the middle panel, (B) prevalence across males and females amongst less than 50 years, and in the right (C)

prevalence amongst both males and females in the higher age group more than or equal to 50 years along the horizontal axis (x-axis) in percent (%) and y axis on the left is the states and union territories in India. In the plot, navy blue represents age group more than 50 years and green represent age group less than 50 years; blue represent proportion of male and red represent proportion of female. Figures have been independently developed from the publicly available database of Global Burden of Disease (GBD) study reference 17. [Accessed on 6 June, 2021 at http://vizhub.healthdata.org/gbd-compare/india]

FIGURE 2B Prevalence of Atopic dermatitis in adults across all states and union territories in India

The left panel show the prevalence (percent) of atopic dermatitis in adults among A) the age group less than 50 years and higher age group, in the middle panel, (B) prevalence across males and females among less than 50 years, and in the right it is (C) prevalence among both male and female in the higher age group more than or equal to 50 years along the horizontal axis (x-axis) in percent (%) and y axis on the left is the states and union territories in India. In the plot, navy blue represents age group more than 50 years and green represent age group less than 50 years; blue represent proportion of male and red represent proportion of female. Figures have been independently developed from the publicly available database of Global Burden of Disease (GBD) study reference 17. [Accessed on 6 June, 2021 at http://vizhub.healthdata.org/gbd-compare/india]

FIGURE 3. Sensitization of A) House Dust Mites (HDM) species B) Pollen allergens (clinical important) and their spatial pattern across India.

From the left, HDM sensitisation rates over the spectrum green to red and the HDM species with symbols in the order listed *Dermatophagoides farinae, Dermatophagoides pteronyssinus, Blomia tropicalis, Acarus siro, Lepidoglyphus destructor, Tyrophagu putrescentiae*, and total HDM. Figure in right represent the different pollen allergens with sensitisation rates over the spectrum green to red with symbols as species names from various clinic-immunological studies across India. The pattern of sensitization to pollens in India across five regions is explained in detail from Bhattacharya *et al*, 2018 (Table 1) and the Pan India distribution of common pollen species/families and allergens discussed from Singh A B and Kumar P, 2003 study in reference [https://doi.org/10.1159/000490805] and [PMID: 14677902] respectively.

FIGURE 4 Risk factors for asthma, allergic rhinitis and atopic dermatitis among children in age group 6-7 years in India

*Odds Ratios (ORs) in the forest plot are highlighted with red show at risk; green shows protective, and black shows no significant association. From the left the

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adjusted odds ratio for A) Asthma B) Allergic Rhinitis, and C) Atopic Dermatitis among children in the age group 6-7 years.

OR1 is the adjusted odds ratio for (N=44,928) and adjustedOR2 is (N=26,403) from ISAAC III and GAN I study India data respectively. The risk factors considered in the studies are: maternal smoking, paternal smoking, sedentariness, weekly exercise, having older siblings, younger siblings, use of paracetamol, outdoor pollution, pollution due to indoor cooking, use of the indoor heating, exposure to moist or damp areas at residence, being male, use of antibiotics, premature birth, extensive breastfeeding, birthweight and measure of BMI as underweight, overweight, and obese.

Figures have been independently developed from the publicly available database of the ISAAC III, India study (2003) from reference 13 [Accessed on 8th May, 2021 at http://doi.org/10.5255/UKDA-SN-8131-1] and GAN phase I, India study (2022) in reference 20 [DOI: https://doi.org/10.1016/j.jacig.2022.01.004]

FIGURE 5 Risk factors for asthma, allergic rhinitis and atopic dermatitis among children in age group 13-14 years in India

*Odds Ratios (ORs) in the forest plot are highlighted with red show at risk; green shows protective, and black show no significant association. From the left the adjusted odds ratio for the A) Asthma B) Allergic Rhinitis and C) Atopic Dermatitis among children in the age group 13-14 years (right panel).

The risk factors considered in the studies are: maternal smoking, paternal smoking, sedentariness, weekly exercise, having older siblings, younger siblings, use of paracetamol, outdoor pollution, pollution due to indoor cooking, use of the indoor heating, use of watercoolers at residence, being male, presence of cats at residence and measure of BMI as underweight, overweight and obese.

OR1 is the adjusted odds ratio for (N=48088) and OR2 is (N=26,403) from ISAAC phaseIII and GAN study, India respectively. Figures have been independently developed from the publicly available database of the ISAAC III study (2003) from reference 13 [Accessed on 8th May, 2021 at http://doi.org/10.5255/UKDA-SN-8131-1] and GAN phase I, India study (2022) in reference 20 [DOI: https://doi.org/10.1016/j.jacig.2022.01.004]

FIGURE 6 Risk factors for asthma among adults in India

† from INSEARCH study (n=14,479) aged ≥35 years and # from GAN study (n=81894) parents of school children. *Odds Ratios (ORs) in the forest plot are highlighted in red show at risk, and in black show no significant association. From the top, the odds ratio for A) Asthma B) Allergic Rhinitis and C) Atopic Dermatitis among adults (bottom panel).

The risk factors studied are sex (woman), living in urban areas, family history of asthma, smoking history of cigarettes, exposure to environmental tobacco smoke

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(ETS), use of traditional method of cooking (coal/wood) for asthma; food allergy (intake of egg), currently smoking tobacco, use of mosquito coils in residence and indoor characteristics (presence of moulds or asthma) for allergic rhinitis and atopic dermatitis respectively.

Figure adapted and modified from the data published from the INSEARCH India study (2012) in reference 4 [DOI: 10.5588/ijtld.12.0005] and GAN phase I, India study (2022) in reference 20 [DOI: https://doi.org/10.1016/j.jacig.2022.01.004]

SUPPORTING INFORMATION

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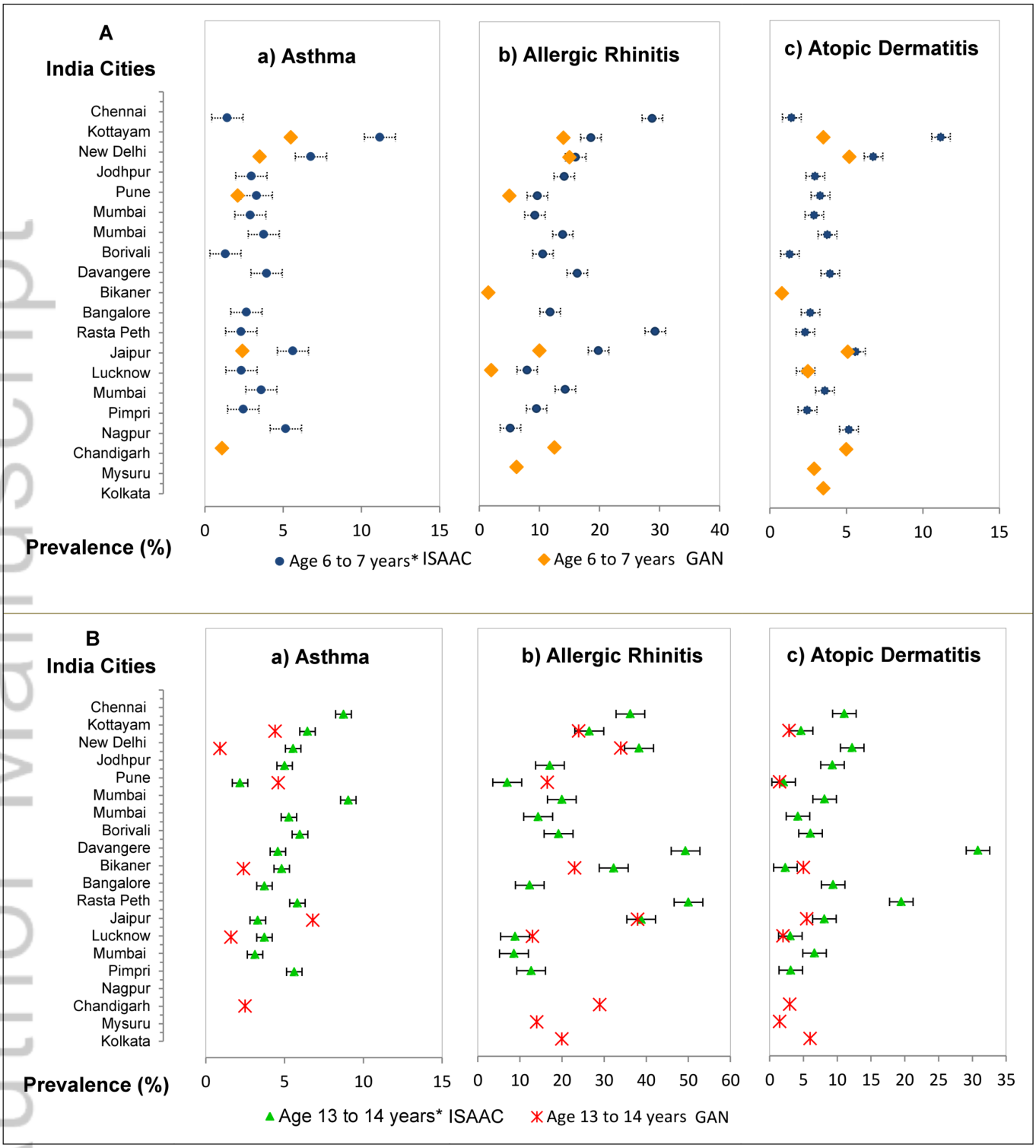

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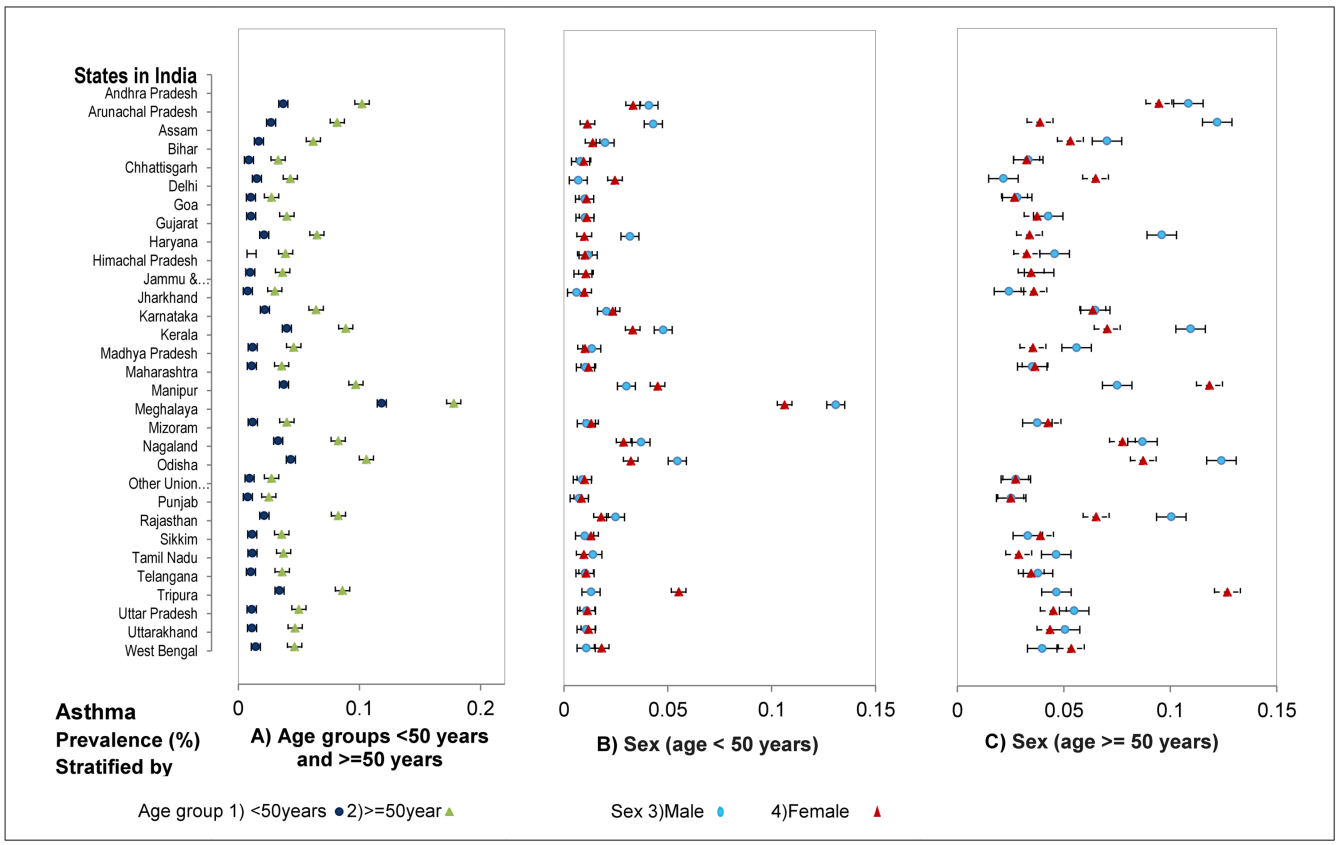

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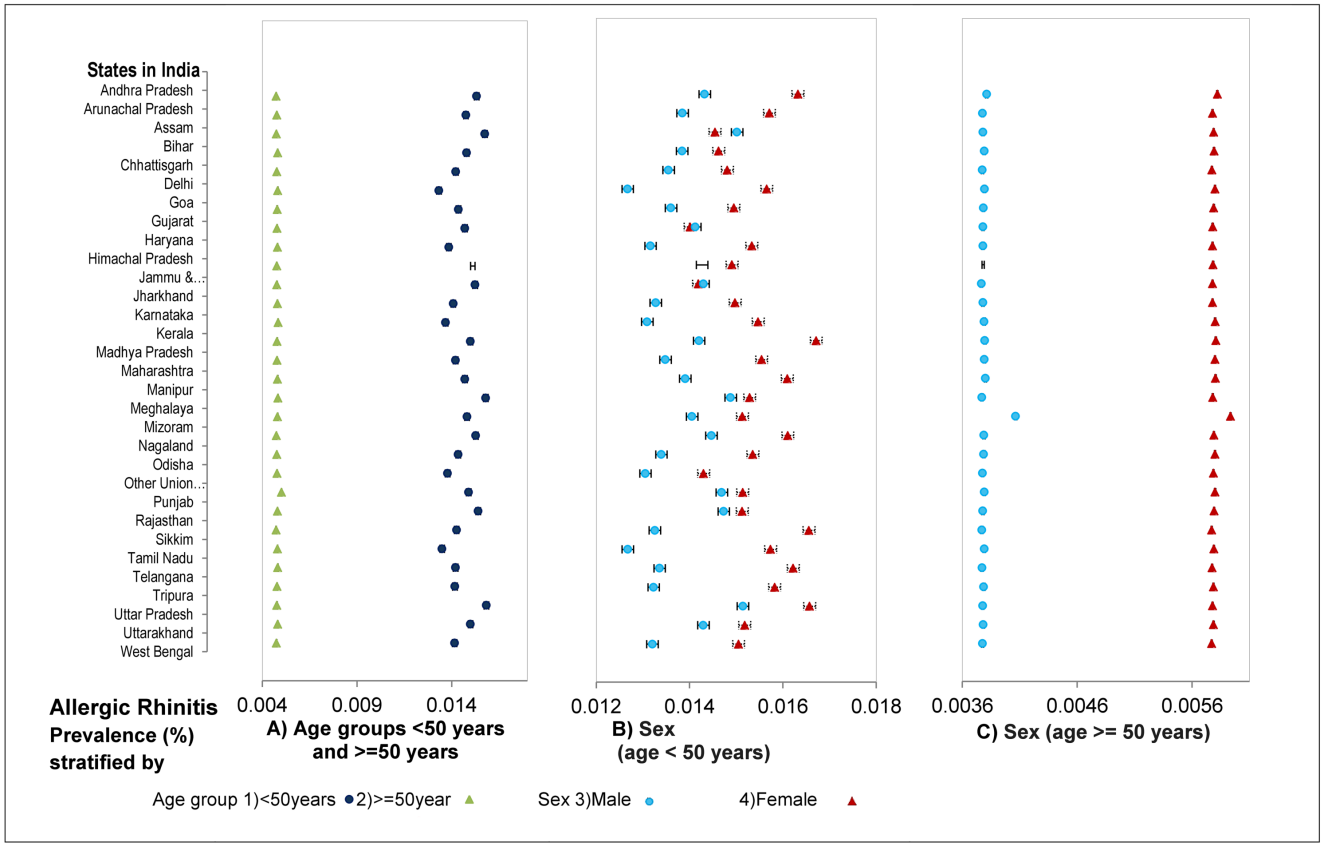

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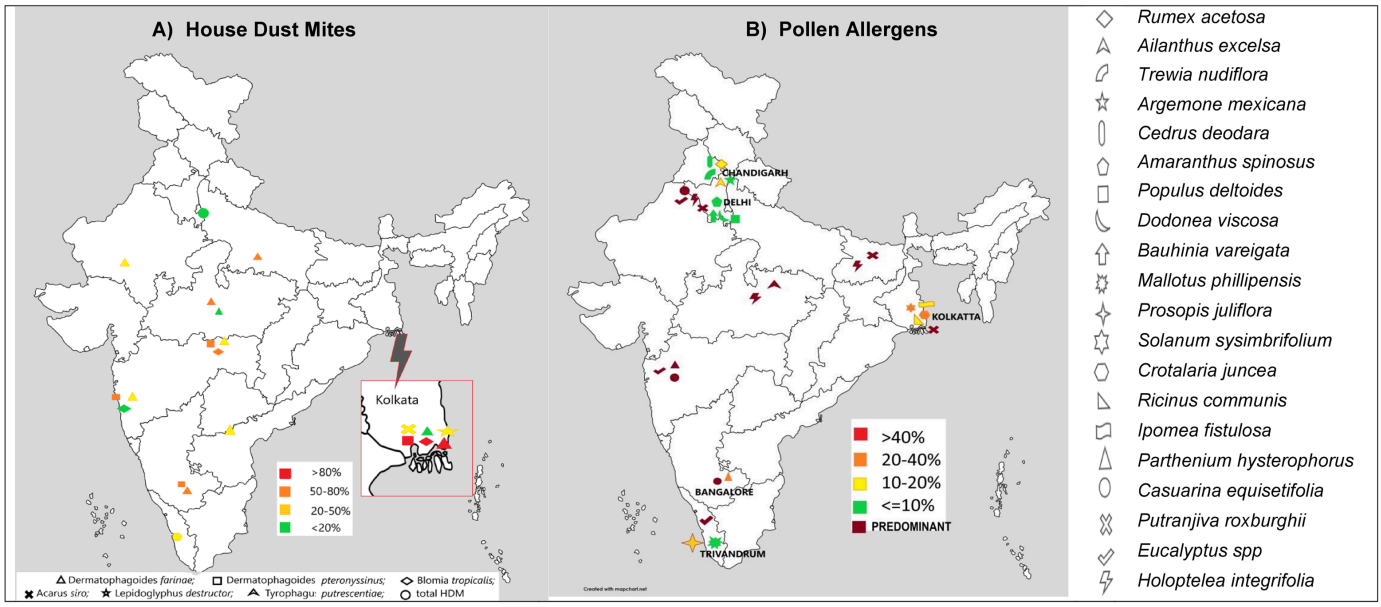

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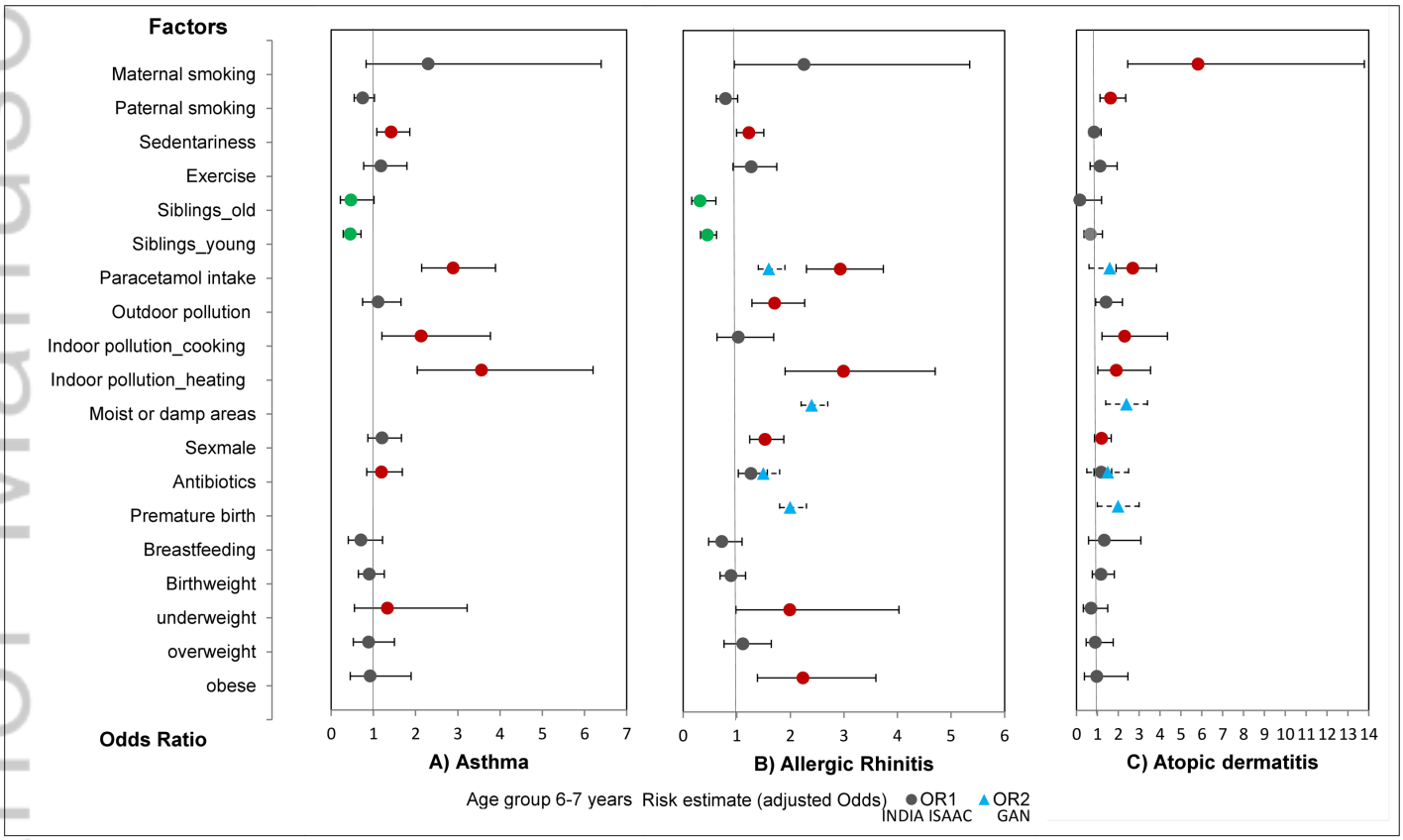

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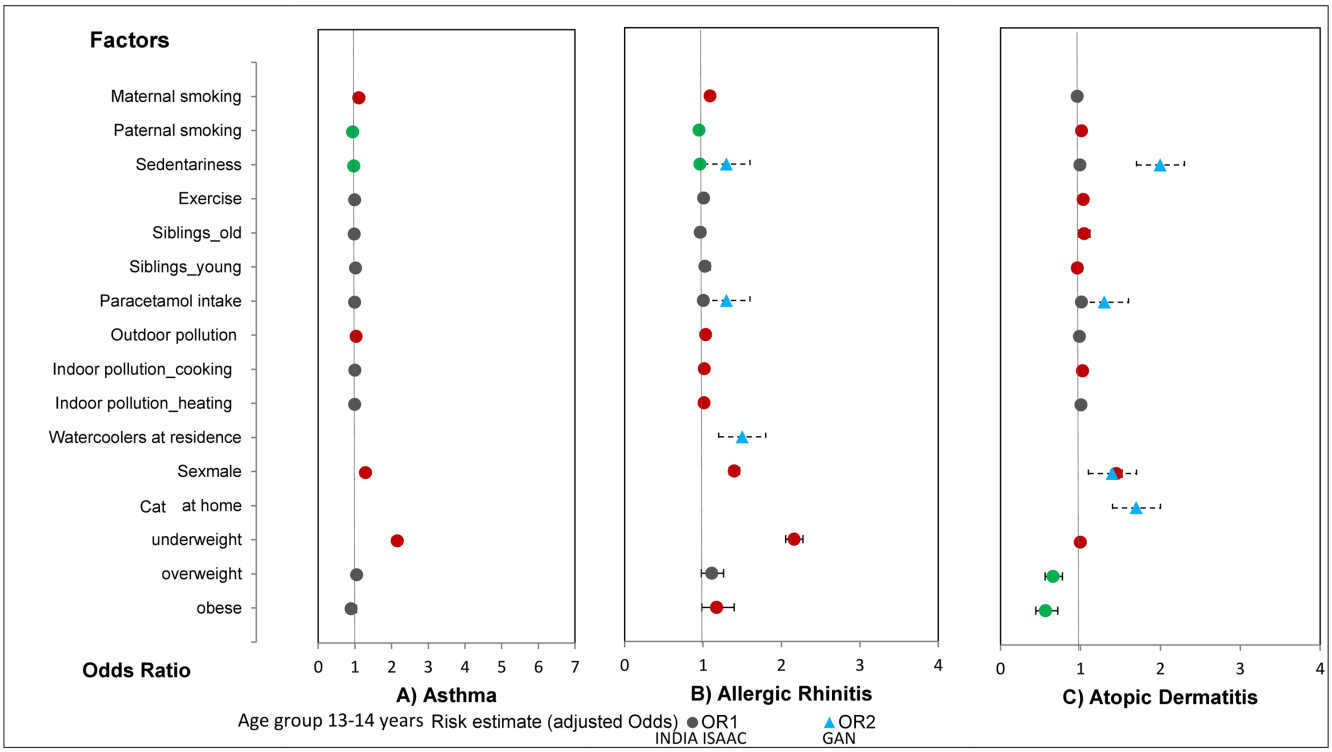

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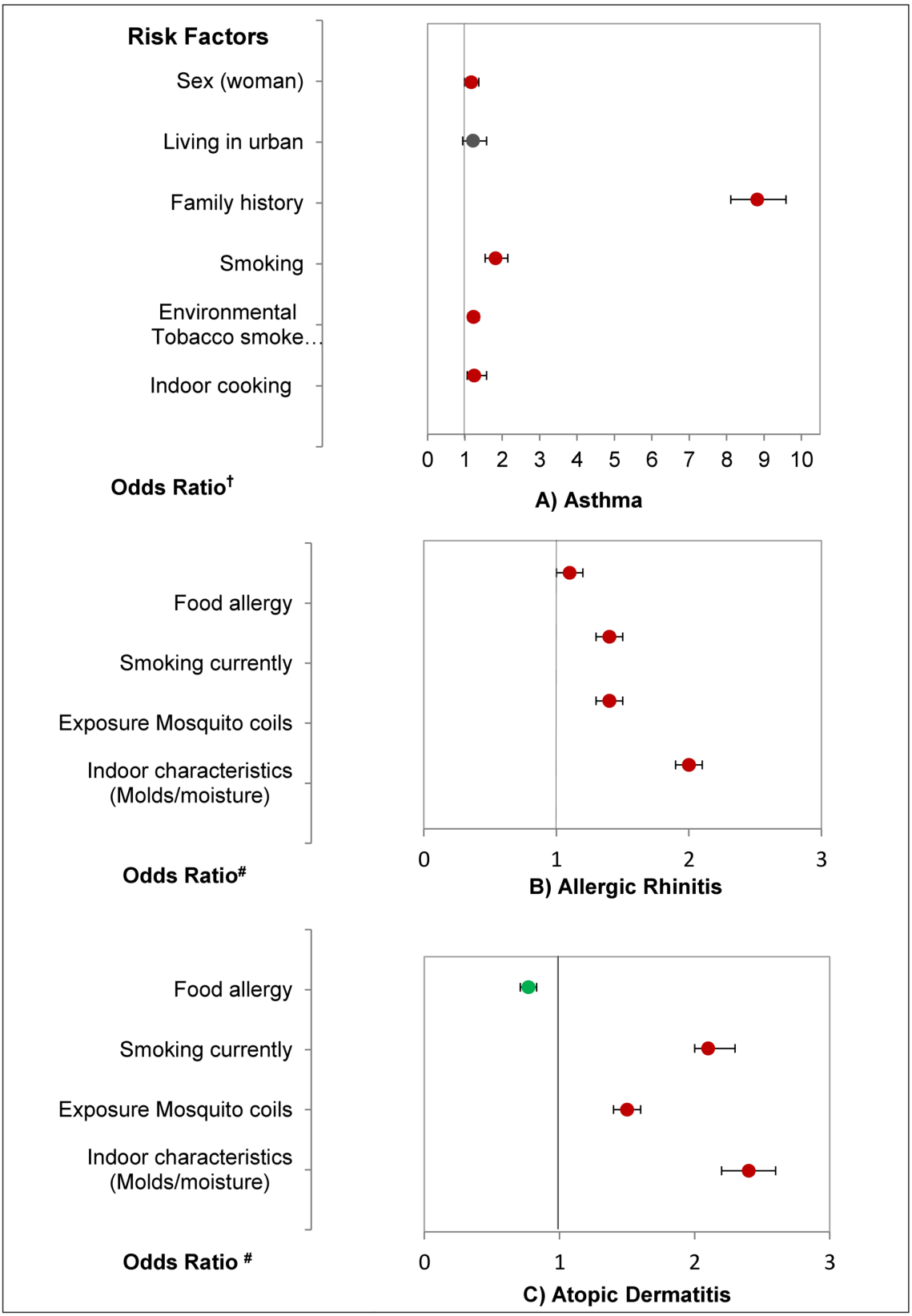

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